Received: 22 May 2024 | Revised: 25 August 2024 | Accepted: 28 August 2024

DOI: 10.1111/cts.70030

ARTICLE

# Computational drug discovery pipelines identify NAMPT as a therapeutic target in neuroendocrine prostate cancer

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## Abstract

Neuroendocrine prostate cancer (NEPC) is an aggressive advanced subtype of prostate cancer that exhibits poor prognosis and broad resistance to therapies. Currently, few treatment options are available, highlighting a need for new therapeutics to help curb the high mortality rates of this disease. We designed a comprehensive drug discovery pipeline that quickly generates drug candidates ready to be tested. Our method estimated patient response to various therapeutics in three independent prostate cancer patient cohorts and selected robust candidate drugs showing high predicted potency in NEPC tumors. Using this pipeline, we nominated NAMPT as a molecular target to effectively treat NEPC tumors. Our in vitro experiments validated the efficacy of NAMPT inhibitors in NEPC cells. Compared with adenocarcinoma LNCaP cells, NAMPT inhibitors induced significantly higher growth inhibition in the NEPC cell line model NCI-H660. Moreover, to further assist clinical development, we implemented a causal feature selection method to detect biomarkers indicative of sensitivity to NAMPT inhibitors. Gene expression modifications of selected biomarkers resulted in changes in sensitivity to NAMPT inhibitors consistent with expectations in NEPC cells. Validation of these markers in an independent prostate cancer patient dataset supported their use to inform clinical efficacy. Our findings pave the way for new treatments to combat pervasive drug resistance and reduce mortality. Furthermore, this research highlights the use of drug sensitivity-related biomarkers to understand mechanisms and potentially indicate clinical efficacy.

## Study Highlights

### WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

Neuroendocrine prostate (NEPC) represents one of the most aggressive subtypes of prostate cancer and has very limited treatment options. Patients affected with NEPC have very poor survival. Therefore, there is an imperative need for effective therapeutics to combat the current high mortality rates.

**WHAT QUESTION DID THIS STUDY ADDRESS?**

Traditional drug research and development pipelines require considerable time and capital input. Computational drug discovery strategies have emerged as a cost-efficient alternative to quickly propose actionable drug predictions. Especially in situations like NEPC, where the affected patient population is relatively small but the need for novel therapeutics is urgent, computational methods can play a pivotal role at providing pharmacological opportunities. In addition, biomarkers indicative of treatment response can expedite clinical development of novel therapeutics. To this end, we employ a novel approach based on causal feature selection that infers direct causes and effects of drug exposure.

**WHAT DOES THIS STUDY ADD TO OUR KNOWLEDGE?**

We have identified NAMPT inhibitors to be efficacious against NEPC and validated their potency in vitro. Our results further add necessity into clinical evaluation of NAMPT inhibitors in NEPC patients. Once validated, they may alleviate the current poor prognosis. Using causal inference, a collection of biomarkers indicative of therapy response was identified and further validated. They may aid patient selection and facilitate clinical investigations of proposed drugs.

**HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY OR TRANSLATIONAL SCIENCE?**

Our findings solidified the use of computational methods to quickly generate drug predictions for specific indications. Biomarkers identified through causal feature selection harbor high predictivity that may benefit clinical development. This comprehensive approach may be adapted to screen for novel therapeutics in other complex diseases.

# INTRODUCTION

Today, prostate cancer (PC) continues to be the most diagnosed malignancy and a leading cause of cancer deaths among US men.[1] Most PC-related deaths occur after onset of therapy resistance, an advanced stage often named castration-resistant prostate cancer (CRPC) due to pervasive unresponsiveness to hormone therapies.[2–4] Though most CRPC tumors remain adenocarcinoma-like (Adeno-CRPC) and dependent on androgen receptor (AR) activity,[5] a considerable proportion of CRPC patients possess small-cell neuroendocrine (NE) properties and lack AR dependency.[6–9] These NE-like CRPC tumors, or neuroendocrine prostate cancer (NEPC), are associated with very poor patient survival because of their intrinsic resistance to hormone therapies and transient response to platinum-based chemotherapies.[10,11] To date, there has been no targeted treatment approved for NEPC, underlining a pressing need to discover therapeutic opportunities to curb the current high mortality rate.

Aided by large-scale molecular profiling techniques, multiple studies have shown divergent transcriptomic patterns between Adeno-CRPC and NEPC patient populations.[6,11–13] These studies often survey genomic factors indicative of differentiation or emergence of NE characteristics. For example, Beltran et al. calibrated a set of genes of which expression differed drastically between Adeno and NE CRPC populations and proposed an activity metric named NEPC score, a molecular phenotype describing a tumor's similarity to typical NEPC tumors.[12] In general, identified biomarkers were used to better understand NEPC biology, rather than to fulfill therapeutic development needs. To search for treatment options, Puca et al.[14] carried out a large-scale drug screen in patient-derived organoids to look for drugs more efficacious in NEPC. Corella et al.[15] identified BCL-2 inhibitors as a drug class for treating NEPC tumors based on elevated expression of BCL-2. Though these studies deliver potential therapeutic options to be further tested, large-scale drug screens in representative models with reproducible results are still lacking. Meanwhile, target identification based on differentially expressed genes may result in critical molecules being skipped due to lack of drugs to target them.

We have previously reported drug discovery efforts using computational screening frameworks that greatly accelerated research and development pipelines for aggressive cancers such as triple-negative breast cancer and CRPC.[16,17] Using high-throughput drug screen (HTS) databases on cancer cell lines (CCLs),[18] we distilled drug–gene relationships and used them to predict responses of patient tumors to various drugs. Meanwhile, patient subtyping based on distinct gene expression patterns can be used to seek drugs targeting specific disease subtypes. Here, given the abundance of CRPC patient molecular profiles, we have designed a computational method to project patient response to various treatments. Utilizing NEPC scores estimated from tumor gene expression data, we have identified various potentially efficacious drug candidates spanning several targets, not only recapitulating previous findings but providing novel treatment strategies. Specifically, we have discovered a class of agents targeting nicotinamide phosphoribosyltransferase (NAMPT) that preferentially inhibits the growth of a NEPC model compared with an Adeno-CRPC model. Our in vitro drug exposure findings further support the potential of NAMPT inhibitors' effectiveness and support our computational drug discovery rationale.

Moreover, to inform clinical development and illuminate the underlying mechanisms of NAMPT inhibitors in NEPC tumors, we developed a biomarker discovery approach to parse key molecules and query their regulatory contributions. We implemented graph-based causal inference algorithms to uncover key genes constituting the most pronounced interactions with NAMPT inhibitors drug sensitivity. These biomarker genes further undergo interrogations in an NEPC-specific context.

Taken together, we report our computational drug discovery and biomarker identification findings for NEPC. Given that phenotypic and genomic assessments of NEPC are becoming ever more prevalent,[19] now is an ideal time to capitalize on in silico methods that leverage big data to quickly provide actionable drug candidate predictions.[20] Our results support adaptations of computational approaches to pursue new therapeutic opportunities for specific diseases or disease subtypes.

# METHODS

## Data acquisition and preprocessing

High-throughput drug screen (HTS) data were downloaded from The Cancer Dependency Map (DepMap, https://depmap.org/portal/). Both the Cancer Therapeutic Response Portal (CTRPv2) and the Genomics of Drug Sensitivity in Cancer (GDSC) screens were obtained through DepMap. The pan-cancer cell line (CCL) transcriptomic profile was downloaded from DepMap, originally from the Cancer Cell Line Encyclopedia (CCLE).[21] CCLE gene expression values in transcripts per million (TPM) were used. In addition, we also downloaded a microarray-measure gene expression profile on CCLs, originally provided by the GDSC. Three CRPC clinical cohort studies with patient transcriptomic profiles were included. The Stand-up-to-Cancer East Coast study (SU2C-EC) was obtained from the cBioportal (https://www.cbioportal.org/, prad_su2c_2019).[22] The Stand-up-to-Cancer West Coast study (SU2C-WC) was obtained from Dr. Scott Dehm, originally generated by Quigley et al.[23] The CRPC-NEPC patient data (Beltran-NEPC) was obtained from Beltran et al.[12] SU2C-EC patient gene expression was available in fragments per kilobase per million (FPKM). SU2C-WC and Beltran-NEPC were both in TPM format. All RNA-seq data were preprocessed through a base-2 logarithm with a pseudocount of 1. All computational analyses were carried out using R (version 4.1.3) provided by the Minnesota Supercomputing Institute (MSI). The R scripts and relevant data are available at the Open Science Foundation (OSF) platform (https://osf.io/jn82k/).

## Projecting drug response in patient tumors

For each CRPC dataset, patient drug response was generated using the R package oncoPredict.[24] First, transcriptomic profiles between a CRPC patient cohort and the CCLE dataset were harmonized to correct batch effects. Genes that were among the least variable (lowest 20 percent) were excluded. Then, for each drug from HTS, a ridge regression model was trained using CCL gene expression values as predictors. Learned coefficients were applied to the CRPC gene expression dataset to generate patient response to corresponding drugs.

## Calculating NEPC scores in CRPC patient cohort data

We designed virtual drug screens to select efficacious treatments for NEPC; that is, selecting drugs that were predicted to be more efficacious among tumors with higher NEPC features. We incorporated the integrated NEPC score metric proposed by Beltran et al.[12] to describe a tumor's proximity to having an NEPC subtype. Briefly, using a set of 70 signature genes, we calculated Pearson's correlation coefficient between a given tumor and a reference NEPC (Table S1). Therefore, a score close to 1 indicates high likelihood of being NEPC. This calculation was applied independently to the three CRPC cohorts to first generate each patient's NEPC score.

## Identifying efficacious drugs for NEPC through statistical modeling

To screen for drug candidates in each dataset, we constructed linear models to examine associations between predicted patient drug response and NEPC scores and to select drugs that are efficacious among tumors with high NEPC likelihood (i.e., high NEPC scores), while adjusting for potential confounders. To increase robustness, we applied this pipeline independently in each of the three CRPC patient datasets. Depending on availability of variables that can inform potential confounding, in SU2C-EC, we established linear models such that

$$\text{NEPC Score} = \beta_0 + \beta_1 \times \text{Drug Response} + \beta_2 \times \text{Exposure} + \beta_3 \times \text{PSA},$$

where Exposure is a binary variable indicating treatment history of ADTs (1 if the participant had received ADTs at biopsy and 0 otherwise). PSA is a continuous variable representing prostate-specific antigen level of CRPC patients. This examines associations between predicted patient response and SU2C-EC tumor NEPC scores, while adjusting for confounders like treatment status and PSA. Similarly, for SU2C-WC, we used

$$\text{NEPC Score} = \beta_0 + \beta_1 \times \text{Drug Response} + \beta_2 \times \text{Survival} + \beta_3 \times \text{PSA},$$

where Survival, a binary variable indicating the occurrence of death during study follow-up (1 if yes and 0 otherwise). It was used as a surrogate confounder for resistance to treatment received, since a patient who displayed worse survival may bear higher resistance to the treatments. This may confound NEPC score because NEPC tumors display high resistance to most therapies. For Beltran-NEPC, we screened drugs via

$$\text{NEPC Score} = \beta_0 + \beta_1 \times \text{Drug Response} + \beta_2 \times \text{Purity},$$

where Purity represents tumor purity (fraction of cancer cells relative to all cells), a value estimated based on a previously reported method.[25] In all three datasets, Drug Response is a continuous variable representing the predicted sensitivity (pharmacological AUC), and the estimated $\beta_1$ indicates associations between drug response and NEPC scores.

After model fit, we selected drugs that had a negative association (coefficient) using a threshold of Benjamini–Hochberg adjusted $p$-value <0.05. In addition, we calculated Pearson's correlation coefficient between drug response and NEPC score and selected drugs whose correlation were lower than −0.4 as a threshold for effect size. Eventually, identified drugs from each patient dataset were pooled, and those that were independently selected in all three datasets were considered as candidates.

## Calculating NEPC scores of CRPC in vitro cell line models

We calculated NEPC scores of potential in vitro CCLs using the same NEPC reference vector. In addition to the CTRP, which includes the known NEPC CCL HCI-H660, we included CCLs from the GDSC, a pair of genetically engineered CRPC CCLs (ENZ-sensitive R1-AD1 and ENZ-resistant R1-D567) from Nyquist et al. (GEO Accession GSE49196), and ENZ-resistant 42D and 42F cells from Davies et al. (GEO Accession GSE138460). Prior to calculating Pearson's correlation coefficient between each CCL and the reference vector, batch effects between different data sources were adjusted using the R package combat.[26]

## Causal feature selection for daporinad sensitivity biomarkers

We implement a graph-based method to infer causal structure (or subgraph) of drug response. The drug response variable, $y$, in this situation is the measured CCL sensitivities to daporinad in AUC. By also inputting CCL gene expression ($X$), the causal feature selection algorithm detects genes ($Z$) that correlate with $y$ even when conditioning on any subset of the remaining genes ($X - Z$). As causal feature selection algorithms aim to detect a small set of variables with strong influences on the response variable, we first screened for genes whose expressions were highly correlated with drug response via the R package limma.[27] Univariate linear models between CCL response to an NAMPT inhibitor and CCL expression of each gene were constructed, through which the top 800 genes with the highest $B$-statistics (log-odds of a gene being differentially expressed) were kept. To identify potential regulatory genes closely related to NMAPT sensitivity, we utilized the PC-simple algorithm[28,29] and input the CCL NAMPT sensitivity again with subsetted CCL expression profiles containing only the 500 genes. A typical output from PC-simple is a set of features that remain correlated with the response variable even when conditioning on any other features or sets of features. To avoid spurious results, we generated 100 bootstrap samples by randomly selecting 80% of CCLs with replacement. Eventually, we consider genes identified from at least 20 bootstrapping repetitions as robust sensitivity biomarkers. To visualize the local causal structure around drug sensitivity, we ran the PC algorithm using a dataset with features including

CCL drug response and the selected biomarker genes from PC-simple. We repeated PC algorithms on 100 randomly generated bootstrap samples to estimate confidence of edges between nodes in this local graph. Adjacency matrices from each calculation were obtained to estimate probability for a given edge between nodes. The eventual local subgraph was visualized using the R package igraph with edge probabilities interpreted as weights.[30]

## Daporinad sensitivity score using expression of biomarker genes

We applied daporinad sensitivity biomarkers independently in CRPC patient datasets through calculating an activity score indicating likelihood of response to daporinad. A given CPRC gene expression dataset was first filtered to contain only identified biomarker genes; then, *z*-scores were calculated for each gene across all samples by normalizing expression values to have a zero mean and a unit standard deviation. For each patient tumor sample, *z*-scores from all biomarkers were added to generate activity scores. These scores are then scaled to vary between 0 and 100, with a higher value indicating higher sensitivity to daporinad. Note that this is a unit-free scaler, different from the drug response statistic, that is, AUC.

## CRPC patient biopsy single-cell RNA-seq analysis

We obtained the CRPC patient biopsy scRNA-seq data from Chan et al.[31] (GEO Accession: GSE210358). Overall, 27,338 cells from 12 patients were included in the dataset. Among them, three patients were categorized as NEPC whereas nine patients have adenocarcinoma phenotypes. To account for data noise at the individual-cell level, we generated pseudobulk samples representing heterogeneity of the patient cohort. Based on the provided cellular clusters, we generated a pseudobulk sample using get_pseudobulk from the python library decoupler.[32] Mean gene counts were aggregated over all available cells within each cluster. Pseudobulks with at least 10 cells and 1000 total counts were retained for DSS calculation. Calculated DSS between NEPC and CRPC-Adeno pseudobulks were tested via a Wilcoxon *t*-test.

## Cell lines and culture reagents

The LNCaP and NCI-H660 cell lines were purchased from ATCC (Manassas, VA, USA). Both cells were maintained at 37°C with 5% $CO_2$ and cultured in RPMI1640 medium with 10% FBS for LNCaP, or HITES medium (RPMI1640 medium plus 0.005 mg/mL Insulin, 0.01 mg/mL Transferrin, 30 nM Sodium selenite, 10 nM Hydrocortisone, 10 nM beta-estradiol and 4 mM L-glutamine) with 5% FBS for H660. All cell lines were periodically monitored for mycoplasma using the Universal Mycoplasma Detection Kit following the manufacturer's protocol (ATCC). The therapeutic compounds (E)-Daporinad (FK866; CAS No. 658084-64-1), ABT-737 (CAS No. 852808-04-9), and GMX-1778 (CHS-828; CAS No. 200484-11-3) were obtained from MedChem Express (Monmouth Junction, NJ, USA) and dissolved in dimethylsulfoxide (DMSO) to standardized stocks at a concentration of 5 mM.

## Generation of stable NAMPT/HNRNPA1 knockdown and NAMPT overexpression lines

Stable knockdown and overexpression models for either NAMPT or HNRNPA1 were created from the NCI-H660 or LNCaP lines utilizing custom designed shRNA and mammalian gene expression vectors obtained from Vector Builder (VectorBuilder Inc., Chicago, IL, USA). A shRNA scramble control and a mCherry expression control vector were utilized as transduction controls. Detailed information for each of the knockdown and expression vectors are provided in Table S2. LNCaP cells were seeded at a density of $1 \times 10^5$ cells per well in a six-well tissue culture plate and allowed to adhere for 24 h in a 37°C humidified incubator with 5% $CO_2$ prior to transduction. LNCaP cells were transduced for 24 h in 1 mL of growth media containing an appropriate amount of lentivirus to obtain an MOI of 10 following the manufacturer's protocol. NCI-H660 cells were plated at a density of $5 \times 10^6$ cells per well in a 6-well tissue culture plate suitable for suspension cells and transduced for 72 h in 2 mL of growth media containing lentivirus and 8 μg/mL polybrene. Following transduction, all knockdown, overexpression, and vector control models underwent a 5-day antibiotic selection with puromycin utilizing a concentration of 2 μg/mL for LNCaP and 0.5 μg/mL for H660. Positive GFP-expression cells were visualized and counted daily during selection to confirm transduction efficiency using the Cytation™ 1 Cell Imaging Multi-Mode Reader (BioTek Instruments, Winooski, VT, USA). Confirmation of target gene expression was performed by assessing NAMPT and HNRNPA1 expression using quantitative real-time PCR.

## Gene expression validation

Total RNA was extracted from cultured cells using the Quick-RNA MiniPrep Plus Kit (Zymo Research, Irvine, CA, USA) and quantified on the NanoDrop ND-8000

8-channel spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA). A total of 2 μg of RNA was used to synthesize cDNA, utilizing the High Capacity cDNA Reverse Transcription Kit (Thermo Fisher Scientific). The Sso-Advanced Universal SYBR Green SuperMix (Bio-Rad, Hercules, CA, USA) was used to conduct real-time PCR analyses following the manufacturer's protocol. Pre-designed primers for all genes were selected from PrimeTime qPCR Primer Assays avail-able from Integrated DNA Technologies, Inc. (IDT; Coralville, IA, USA). The PCR primer sequences utilized to assess gene expression are provided in Table S3. Quantitative real-time PCR and data collection were performed using the 7500 Real-Time PCR System (Applied Biosystems, Foster City, CA, USA). All results were normalized with the expression of GUSB and HPRT1 as a reference panel. All PCR reactions were performed in triplicate, and each knockdown and overexpression experiment was performed independently two times. Expression results were quantified using the ΔΔCt method relative to the appropriate control.[33]

## Protein quantification following gene expression modification

Whole cell lysates of our cell models were collected using RIPA Lysis and Extraction Buffer (Thermo Scientific) containing protease and phosphatase inhibitor (Thermo Scientific) dissolved within. Also, 20 μg of total protein was loaded per lane onto a 4%–20% Mini-PROTEAN® TGX™ Precast Protein Gel (BIO-RAD) and electrophoresis was run at 90 volts for 90 min. Dry transfer onto a PVDF membrane was then performed using an iBlot™ 2 transfer system and its corresponding transfer stacks (Thermo Scientific). Monoclonal antibodies used were NAMPT (NAMPT Rabbit mAb; 56 kDa; A22044, ABclonal), HNRNPA1 (HNRNPA1 Rabbit mAb; 39 kDa; A18094, ABclonal), and GAPDH (GAPDH Rabbit mAb; 36 kDa; A19056, ABclonal) as a loading control. Imaging of blots was conducted using horseradish peroxidase-conjugated secondary antibody on a LI-COR Odyssey Fc imager. Image processing was analyzed with ImageJ.

## Drug screening and viability assays

Cell growth inhibition following drug exposure experiments for all cell line models was assessed using the cell proliferation reagent WST-1 (CellPro-Ro; Roche Applied Science) following the manufacturer's protocol. Briefly, cells were seeded at $1 \times 10^4$ cells per well of a 96-well plate (Thermo Scientific). After 24 h, the cells were then treated with serial dilutions of the individual tested compounds at concentrations ranging from 0 (DMSO control), 0.25 to 64 μM. Cell growth inhibition was measured at 72 h after treatment by incubation at 37°C and 5% $CO_2$ with the WST-1 reagent for 2 h, followed by measuring absorbance at the 450 nm wavelength using the Synergy HTX Multi-Mode Plate Reader (BioTek). Cell viability was determined by normalizing raw absorbance readings to the median values of the no drug treatment control (DMSO) on a per-plate basis. All cell growth inhibition results are reported as a mean and standard deviation of three independent biological experiments, each containing three technical replicates for each experimental condition.

## Statistical analysis

For each growth inhibition experiment, percent viabilities under different doses were used to fit a four-parameter log-logistic regression model using the R package drc.[34] Estimation and the 95% confidence interval (C.I.) are obtained from the resulting dose–response curves. To compare response between CCLs (e.g., LNCaP vs. NCI-H660) or between conditions (e.g., KD vs. Control), the function compParm was is used to test differences in IC50s between the strata and generate a $p$-value with a type-I error rate of 0.05.

# RESULTS

## Virtual drug screen platform for NEPC

Figure 1 describes our computational framework for identifying candidate drugs to treat NEPC. We first generated predictions of drug response in CRPC tumor samples using a previously described method.[24] Briefly, ridge regression models were trained to learn drug–gene associations using CCL drug response and transcriptomic profiles. Learned coefficients were then applied to CRPC patient tumors to predict their sensitivities to the drug. Meanwhile, based on expression patterns of NEPC signature genes, an NEPC score was calculated for each CRPC patient tumor to reflect its likelihood of having a typical NEPC molecular phenotype. Next, we associated predicted patient drug response with NEPC scores via linear models while controlling for potential confounders (see Methods). To select drug candidates, we implemented a threshold to filter for negative drug coefficients with high confidence such that tumors with higher NEPC scores show higher predicted sensitivity (lower area-under-the-dose–response-curve, or AUC) to the drugs.

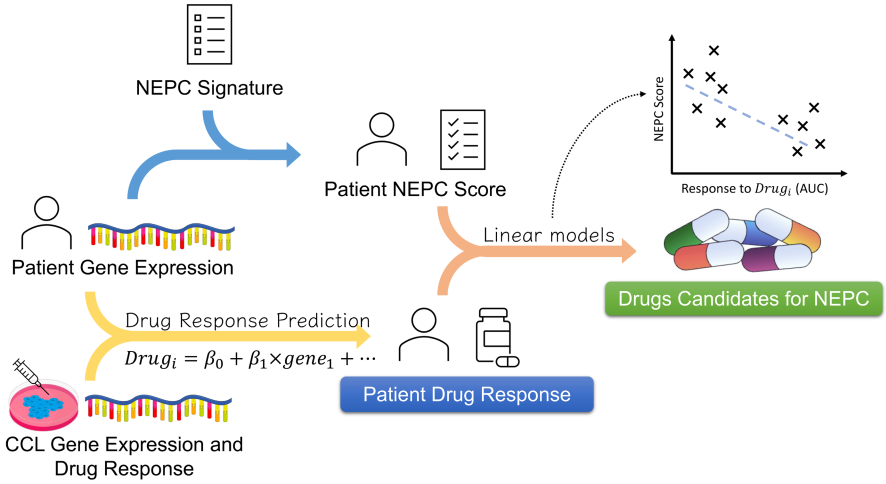


**FIGURE 1** Drug discovery pipeline for NEPC. Our computational drug screen pipeline first projects castration-resistant prostate cancer patient response to various drugs using transcriptomic profiles and HTS on CCLs. Each patient sample's resemblance to NEPC is described by an NEPC score. We construct linear models to examine associations between NEPC scores and predicted patient drug response, while adjusting for potential confounders (see Methods). Drugs projected to have high efficacy among tumors with high NEPC scores (negative correlations between predicted AUC and NEPC scores) are kept for downstream analysis. AUC, area-under-the-dose–response-curve; CCLs, cancer cell lines; HTS, high-throughput drug screens; NEPC, neuroendocrine prostate cancer.

## Diverse drugs show robust efficacy against NEPC

We applied the NEPC drug discovery pipeline described in Figure 1 independently in three CRPC clinical cohort datasets, namely SU2C-EC,[35] SU2C-WC,[23] and Beltran-NEPC.[12] Utilizing the Cancer Therapeutic Response Portal v2 (CTRPv2) for input of drug response measurements on CCLs, for each CRPC patient dataset, sensitivities to 493 treatments were generated. Meanwhile, we calculated an NEPC score for each tumor based on its gene expression concordance with a reference NEPC profile containing specific NEPC signature genes (Table S1).[12] Across SU2C-EC and Beltran-NEPC where histological assertion of NE features was available, NEPC tumors show significantly higher NEPC scores (Figure S1). Next, predicted patient drug response and NEPC scores were associated via linear models, from which drugs were selected based on showing confidently higher predicted efficacy among patients with high NEPC scores (see Methods). We have also included potential variables that may confound NEPC scores into these linear models, given that NEPC is a complex disease with many associating clinical features. First, NEPC patients tend to have low prostate-specific antigen (PSA) levels compared to Adeno-CRPC.[19] Also, many NEPC cases develop with intense treatment of ADTs,[36] and there is a possibility that the patients who had received ADTs possessed higher likelihood of NEPC. NEPC is known to have very poor survival, and patients who had deceased during study follow-up may have had NE features. Therefore, we included PSA, exposure to ADTs, and survival as covariates whenever applicable, given the availability of these variables in each dataset. Since we calculated NEPC scores based on RNA-seq, tumor purity may affect accuracy of NEPC scores if a biopsy contained higher content of non-cancer cells. Therefore, we also included tumor purity as a potential covariate in the Beltran-NEPC dataset. The inclusion of these statistical models mainly enables identifying differential tumor drug response with varying NEPC scores, meanwhile adjusting for potential covariates related to NEPC scores. Therefore, we do not impose predictive or causal interpretations for these models. To obtain robust drug candidates, the drugs identified independently across all three CRPC cohorts were prioritized. In total, 29 drugs were found to show higher efficacy against NEPC tumors across all three independent analyses (Figure 2a). Identified drug candidates were summarized based on their main known molecular targets provided by the CTRPv2 database (Table S4). We then ranked the molecular targets based on the numbers of targeting drugs that were nominated (Figure 2b). Among all robust drug candidates, BCL-2 inhibitors and NAMPT inhibitors both showed strong signals as multiple drug candidates target the same molecules. In SU2C-EC data, projected patient response to ABT-737, a BCL-2 inhibitor, showed a significant correlation with NEPC score (Pearson's $R = -0.56$; $p$-value $< 0.0001$, Figure 2c). Previously, Corella et al. studied systematic gene expression shifts in NEPC compared with AR-active PC and proposed BCL-2 as a druggable target for its consistent upregulation among NEPC samples.[15] They further established extensive experimental evidence that BCL-2

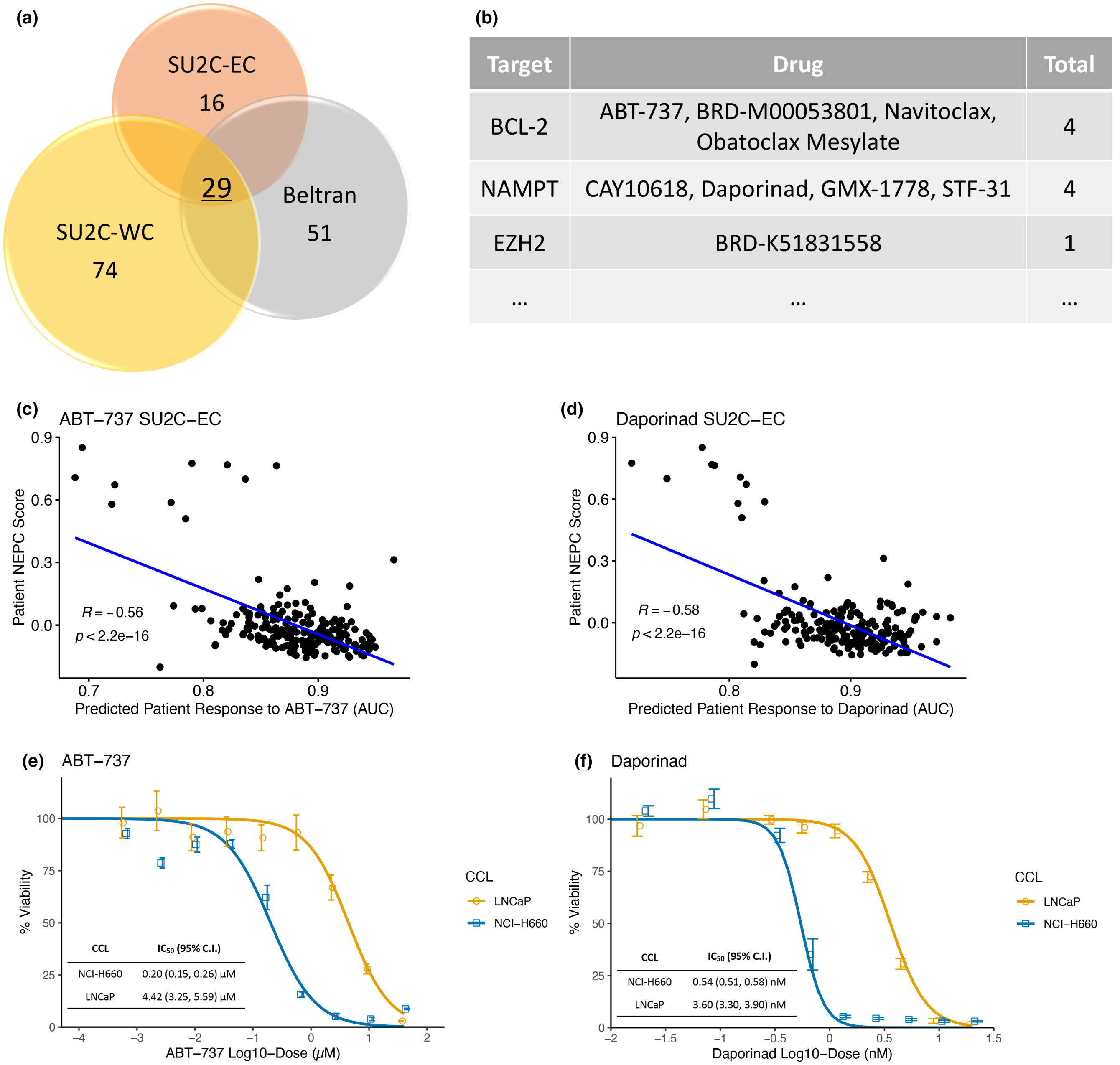


**FIGURE 2** Robust drug discovery pipelines identify diverse candidate drugs to treat NEPC. (a) 29 drugs were identified independently across all three CRPC cohorts. (b) Candidate drugs were summarized based on their molecular targets; drug targets were ranked based on the numbers of targeting drugs. (c) Correlation between predicted sensitivity to ABT-737 and NEPC scores in the SU2C-EC cohort. Pearson's correlation coefficient is shown. (d) Correlation between predicted sensitivity to daporinad and NEPC scores. Complete scatterplots showing associations between predicted drug response and NEPC scores in all three clinical studies can be found in Figure S2. (e) Growth inhibition curves of NCI-H660 and LNCaP cells treated with ABT-737. (f) Growth inhibition curves of NCI-H660 and LNCaP cells when exposed to daporinad. For drug response curves in (e) and (f), mean percent viability at each concentration was plotted along with error bars indicating its standard errors ($N=8$ for ABT-737 screens and $N=6$ for daporinad screens). CRPC, castration-resistant prostate cancer; NEPC, neuroendocrine prostate cancer.

inhibitors induced growth inhibition both in vitro and in vivo, where AR-active samples were less sensitive. NEPC scores in SU2C-EC display strong negative correlations (Pearson's $R=-0.56$; $p<0.0001$) with predicted patient response to daporinad, an NAMPT inhibitor (Figure 2d). This relationship between high predicted sensitivity to NAMPT inhibitors and high NEPC scores was identified independently in all three CRPC cohorts (Figure S2). Knowing that BCL-2 inhibitors have garnered clinical investigation and are currently being evaluated in metastatic CRPC patients,[37] we chose to focus on NAMPT inhibitors because they displayed an equally strong signal resulting from our pipelines and have not prompted substantial clinical development in NEPC.

## Evaluating NAMPT inhibitors in vitro

We next designed in vitro experiments to evaluate efficacy of NAMPT inhibitors. Given our rationale in computational drug discovery, we expected NEPC models to be more sensitive to these drugs compared with Adeno-CRPC models. To establish an in vitro cell line model system that enables such investigation, we obtained comprehensive expression profiles of various PC CCLs from various sources (see Methods). Similar to the CRPC patient data, we calculated NEPC scores for every PC CCL to probe their molecular phenotype (Figure S3A). Unsurprisingly, the known NEPC CCL, NCI-H660, had the highest NEPC score (>0.75) among all available CCLs,[38] and the known AR-reliant cell line LNCaP showed a score close to zero.[39] Therefore, we included NCI-H660 as the NEPC model and LNCaP as the Adeno-CRPC model in our experiment settings.

As a proof-of-concept, we first exposed both NCI-H660 and LNCaP cells to ABT-737, a BCL-2 inhibitor known to selectively inhibit NEPC tumors (Figure 2e). Using a wide range of treatment concentrations, we generated growth inhibition curves for each CCL model. To test drug efficacy differences between CCLs, Student's *t*-tests comparing estimated half-maximal inhibitory concentrations ($IC_{50}$s) from each dose–response curve were implemented (see Methods). Consistent with our computational modeling results and previous findings from Corella et al., ABT-737 induced growth inhibition at low doses in NCI-H660 cells with an $IC_{50}$ of 0.20 μM (95% C.I.: 0.15–0.26 μM), whereas in LNCaP cells the IC50 was over 20-fold higher ($IC_{50} = 4.42$ μM, 95% C.I.: 3.25–5.59 μM, $p < 0.0001$, Table S5). When exposed to the NAMPT inhibitor, daporinad, the NEPC model NCI-H660 had an estimated $IC_{50}$ of 0.54 nM (95% C.I.: 0.51–0.58 nM) while the Adeno-CRPC model LNCaP exhibited a decreased sensitivity with an estimated $IC_{50}$ of 3.60 nM (95% C.I.: 3.30–3.90 nM) (Figure 2f; $p < 0.0001$). Comparable trends were observed for growth inhibition curves induced by another NAMPT inhibitor GMX-1778, in which proliferation of NCI-H660 cells were inhibited at an estimated $IC_{50}$ of 0.62 nM compared to 1.99 nM for LNCaP cells (Figure S3C; $p < 0.0001$). Taken together, these in vitro experiments showing differential NAMPT inhibitor response phenotypes between NEPC and Adeno-CRPC models validate our computational drug discovery rationale. They further provide preclinical evidence for NAMPT inhibitors as an efficacious therapeutic opportunity for treating NEPC.

## Drug response biomarker identification for NAMPT inhibitors

Several previous investigations have revealed higher expression of NAMPT in prostate tumors compared with normal tissues.[40–42] In several neuroendocrine tumor samples from pancreatic cancer and small cell lung cancer donors, inhibiting NAMPT disrupted tumor metabolism and growth.[43,44] These findings justify further inquiries into NAMPT sensitivity related markers in NEPC. To parse molecules associated with differential responses of NEPC and Adeno-CRPC to NAMPT inhibitors, we sought to identify biomarkers for NAMPT sensitivity and further study their pharmacological effects in NEPC. Therefore, we adapted a graph-based causal feature selection algorithm named PC-simple to detect a parsimonious set of influential biomarkers.[28,29] For a causal Bayesian network, PC-simple was developed to discover local relationships, or immediate edges surrounding a target variable.[45] In this scenario, we aimed to identify all the direct causes and effects of NAMPT drug response from transcriptomic profiles through implementation of the PC-simple algorithm (Figure 3a). Since causal discovery algorithms typically test independence between a pair of variables while conditional on various other variables or sets of variables, inputting a high-dimensional dataset without prescreening can be computationally demanding. Thus, using measured CCL sensitivity to daporinad as the target variable, we first identified genes exhibiting high associations with drug response through the R package limma.[27] The top 500 genes ranked by *B*-statistics were further selected for causal inference (see Methods). To obtain robust biomarker genes, we repeated the PC-simple computation in 100 independent bootstrap samples, through which genes identified with a frequency of at least 20 were considered as potential biomarkers.

In total, nine genes were identified with high confidence to be causally related to daporinad sensitivity (Figure 3b). None of the NAMPT inhibitors biomarkers were included in NEPC scoring (Table S1). Immediate interplays among drug response and biomarker genes were visualized through estimated edges in the local network surrounding the target variable (Figure 3c, see Methods). Here, bootstrapping with 200 random samples was implemented to assess confidence in detected causal structures, and edges with a detection probability of at least 0.5 were preserved in the subgraph. Based on the local network, five edges were established between daporinad sensitivity and marker genes including NAMPT, ADAM5, HNRNPA1, PLAAT3, and RIMS3. Among these immediate relationships, NAMPT and RIMS3 both have undirected or bilateral edges with the target variable. The other three genes, ADAM5, HNRNPA1, and PLAAT3, were estimated as the

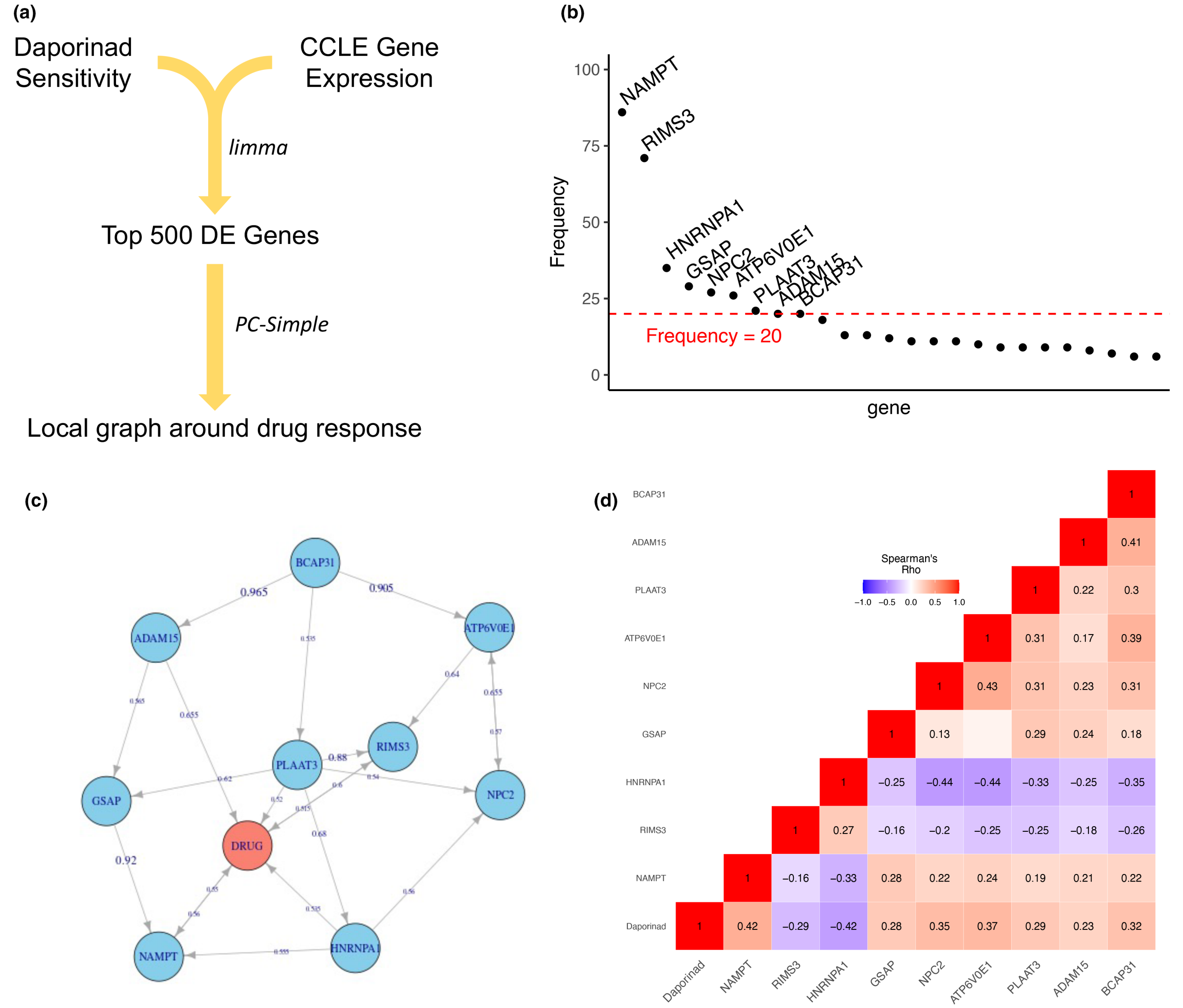


**FIGURE 3** Biomarker discovery for NAMPT inhibitor sensitivity in CCLs. (a) Overview of NAMPT inhibitors biomarker discovery pipeline. (b) Frequencies of causal genes related to measured daporinad sensitivity detected with PC-simple using 100 bootstrapping samples. The red dotted line indicates a threshold frequency of 20. (c) Local causal structure of biomarker genes (blue nodes) and daporinad sensitivity (red node). Each edge is labeled with an estimation of confidence through 200 bootstrap calculations using the PC algorithm. (d) Heatmap showing Spearman's correlation coefficients between each pair of variables in the causal structure depicted in (c). Correlations with a *p*-value <0.05 were labeled. CCLs, cancer cell lines; DE, differentially expressed; NAMPT, nicotinamide phosphoribosyltransferase.

parents of daporinad sensitivity, indicating an upstream regulatory role. A detailed summary of biomarker genes and their known roles in cancer is presented in Table S6.

As expected, NAMPT, the target of daporinad, was chosen most frequently among 85% of bootstrapping repetitions, indicating a direct linkage between NAMPT expression and drug response phenotypes. Based on pairwise Spearman's correlation coefficients between causal variables within this local subgraph, a higher NAMPT expression corresponded to higher response AUC (or greater resistance) to daporinad (Figure 3d). Further examination of NAMPT expression levels between NCI-H660 and LNCaP cells recapitulated this trend (Figure S4), with NAMPT levels in NCI-H660 cells being approximately 57% lower than in LNCaP cells. In line with our observations, a past study showed that depletion of NAMPT in cancer cells caused phenotypic changes that resembled pharmacological intervention.[46] Meanwhile, NAD—the catalytic product of NAMPT—is known as a mechanistic dependency of neuroendocrine tumors,[43] justifying a deeper inspection of its regulatory properties in NEPC.

In addition, another biomarker gene, namely HNRNPA1, has gained continuous scrutiny for its role in prostate cancer, especially its effects on AR protein regulation.[47,48] While residing in the close causal neighborhood around daporinad sensitivity, HNRNPA1 displays an opposite association with the drug response wherein higher levels of HNRNPA1 were associated with lower response AUC (or greater sensitivity) (Figure 3d). Between the two CCLs in our model system, we also observed a borderline significant higher expression of HNRNPA1 in the NEPC CCL NCI-H660 compared to LNCaP (Figure S4). Therefore, illuminating HNRNPA1's modulatory actions in NEPC could add valuable insights for developing NAMPT inhibitors as a treatment option.

## Validation of NAMPT inhibitors sensitivity biomarkers in NEPC

Given the set of NAMPT inhibitor biomarker genes, we next queried their predictability of drug response in NEPC tumors. First, we used identified biomarker genes to calculate a daporinad sensitivity score (DSS) that depicts the magnitude of sensitivity to daporinad (see Methods). Briefly, in a given transcriptomic profile, biomarker expressions were summarized per sample into a unit-free scaler indicative of sensitivity to daporinad. Here, unlike drug response statistics, a higher DSS simply implies higher aggregate levels of the signature set, which corresponds to higher sensitivity to daporinad. As a direct evaluation of DSS predictability, we calculated CCL DSS and correlated the resulting values with measured CCL response to several NAMPT inhibitors from the CTRP, including daporinad, GMX-1778, and STF-31 (Figure S5). Meanwhile, we utilized the Cancer Dependency Map to perform systematic gene association analysis examining CCL gene expression and sensitivity to these NAMPT inhibitors. Statistics of single-gene correlation with drug response were ranked by effect sizes (Table S7). Taken together, DSS displayed strongest association with sensitivity to daporinad ($|r| = 0.68$; $p < 0.0001$), surpassing all single-gene correlation coefficients (maximum $|r| < 0.4$). Moreover, DSS also showed superiority at predicting CCL response to GMX-1778 and STF-31 than any single-gene, suggesting its utility of informing not only daporinad but other NAMPT inhibitor sensitivities.

Note that identification of these causal biomarkers was performed using gene expression and phenotype measurements from CCLs only, without any stratification for prostate cancer origins or involvement of CRPC patient datasets. To evaluate relationships between DSS and neuroendocrine phenotypes, we correlated DSS with NEPC scores in three CRPC clinical cohorts (Figure S6). Across all three CRPC patient datasets, DSS displayed significant positive correlations with tumor NEPC scores, where tumors bearing higher NE features were found to have higher DSS, or more sensitive to daporinad. To further validate DSS in a previously unseen data, we included a single-cell RNA-seq (scRNA-seq) dataset of CRPC patient biopsies.[31] To account for noise at the cellular level, we generated a pseudobulk sample for each single-cell cluster[32] (see Methods). Briefly, each pseudobulk sample aggregated cellular gene expression by obtaining the mean expression levels across all cells in the cluster. Thus, the resulting pseudobulks were less affected by data noise and still represented tumor heterogeneity. In total, 18 NEPC and 38 CRPC-Adeno pseudobulks were generated (Figure 4a). We applied DSS calculation to the pseudobulk dataset and observed that NEPC pseudobulks showed significantly higher DSS compared to CRPC-Adeno pseudobulks (Figure 4b; Wilcoxon $t$-test $p < 0.0001$). To further confirm the robustness of observed DSS difference, we conducted random perturbation experiments by repetitively selecting a random set of genes and calculating an activity score based on this random signature set. Calculated scores between NEPC and CRPC-Adeno pseudobulks were compared using a Wilcoxon $t$-test in a similar fashion. This process was repeated 1000 times, and the resulting test $p$-values were presented in Figure S7. In comparison, DSS separated the two phenotypes by a more significant margin than all of the random perturbations, further validating robustness of DSS against random noise. Thus far, this external validation further substantiates our findings that NMAPT sensitivity-related genes may be informative of treatment effects in NEPC.

Because these biomarkers retain causal relationships with daporinad sensitivity (Figure 3c), we hypothesized that altering their expression levels would alter the sensitivity of NEPC cells to daporinad. To test this, we designed gene expression manipulation experiments in NCI-H660 cells and assayed their response to daporinad under different conditions. As outlined in Figure 4c, we established NAMPT knockdown (KD) and overexpression (OE) NCI-H660 models and a HNRNPA1 KD NCI-H660 model (see Methods). The KD and OE of these molecules were further validated at the RNA as well as the protein levels (Figure S8). Since NAMPT expression positively correlated with daporinad AUC (Figure 3d), we anticipated that NAMPT KD cells would become more vulnerable to daporinad (corresponding to a lower AUC), whereas in NAMPT OE cells would become less vulnerable. Conversely, because HNRNPA1 associates with an elevated AUC, we anticipated that HNRNPA1 KD cells would become less vulnerable. Differences between drug efficacies were tested using IC50 values from fitted dose–response curves via Student's $t$-tests. Consistent with our speculations, inhibiting NAMPT further sensitized NCI-H660 cells to daporinad ($IC_{50} = 0.38$ nM, 95% C.I.: 0.35–0.41 nM) compared

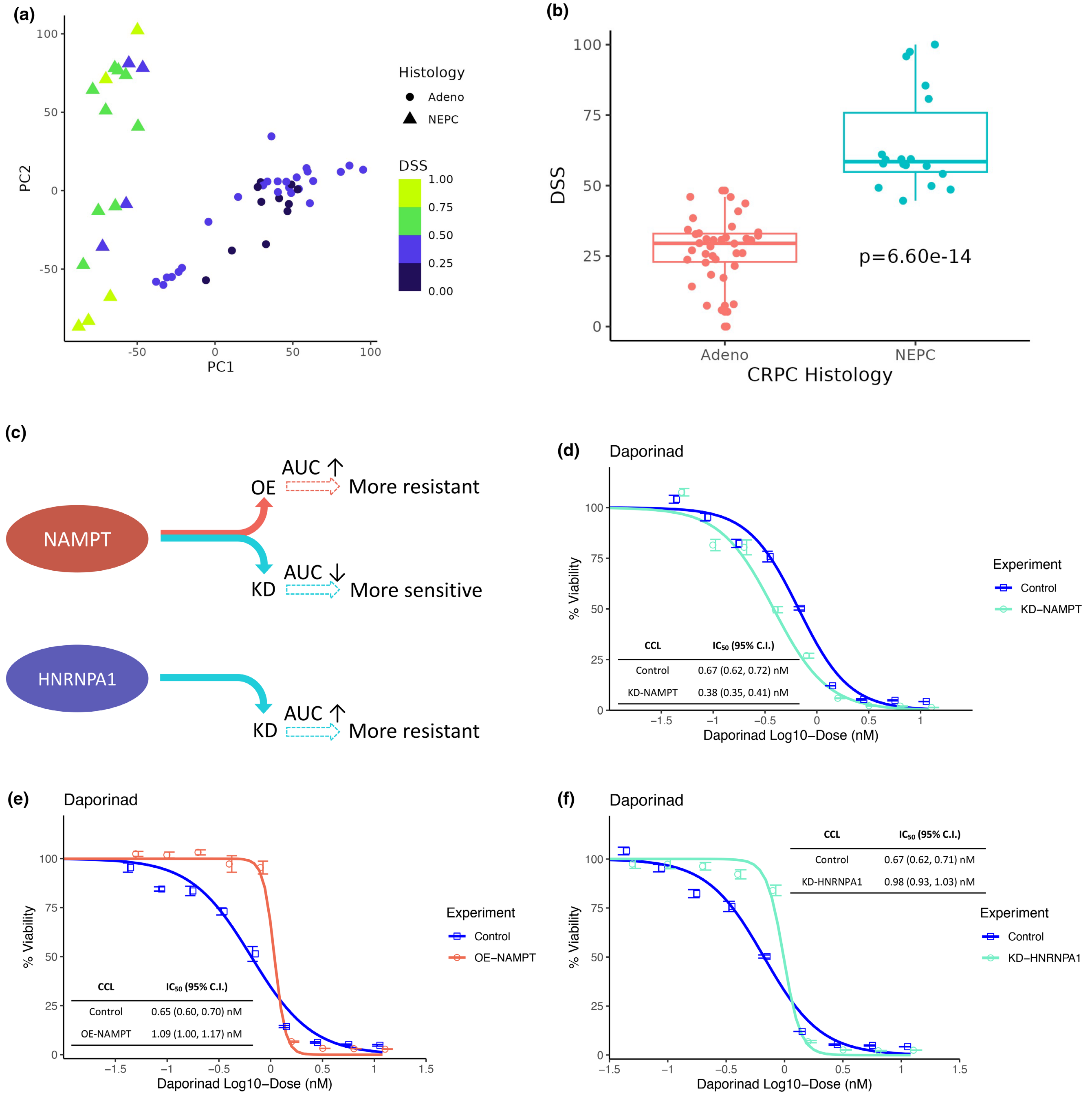


**FIGURE 4** Daporinad biomarkers indicate anti-tumor activities in NEPC. (a) The first two principal components (PCs) of the pseudobulks generated from Chan et al. DSS is a unit-free metric based on expression of daporinad biomarkers. A higher value indicates higher sensitivity to daporinad. (b) Boxplot showing the difference in DSS between NEPC and CRPC-Adeno pseudobulks. Wilcoxon *t*-test *p*-value is provided. (c) Overview of gene expression modifications and their expected effects on sensitivity to daporinad. (d) Growth inhibition curves of NAMPT KD and Control NCI-H660 cells treated with daporinad. (e) Growth inhibition curves of NAMPT OE and Control NCI-H660 cells treated with daporinad. (f) Growth inhibition curves of HNRNPA1 KD and Control NCI-H660 cells treated with daporinad. In (d), (e), and (f), mean percent viability at each concentration was plotted along with error bars indicating its standard errors. AUC, area-under-the-dose–response-curve; KD, Knock-down; NAMPT, nicotinamide phosphoribosyltransferase; NEPC, neuroendocrine prostate cancer; OE, overexpression.

to unmodified control ($IC_{50}$ = 0.67 nM, 95% C.I.: 0.62–0.72 nM) ($p < 0.0001$, Figure 4d). Meanwhile, NAMPT OE reduced sensitivity to daporinad ($IC_{50}$ = 1.09 nM, 95% C.I.: 1.00–1.17 nM) compared to the baseline cells ($IC_{50}$ = 0.65 nM, 95% C.I.: 0.60–0.70 nM) ($p < 0.0001$, Figure 4e). Similarly, HNRNPA1 KD cells display reduced sensitivity to daporinad ($IC_{50}$ = 0.98 nM, 95% C.I.: 0.93–1.03 nM, $p < 0.0001$, Figure 4f). Taken together, our drug exposure findings in gene expression-modified NEPC cells not only confirm NAMPT and HNRNPA1 as predictive biomarkers for daporinad sensitivity in NCI-H660 cells but also reiterate their statistical associations with the response to NAMPT inhibitors. These biomarkers have the potential to inform therapy responses in NEPC patients and might provide insights into mechanisms underlying clinical benefit.

## DISCUSSION

Given the high mortality rate and current lack of treatment options, therapy development for NEPC continues to be an investigational focus.[49–51] Due to similarities of NEPC clinical, molecular, and pathological features with small cell lung cancer, many patients with NEPC are treated with platinum-based compounds. However, these therapies tend to fail after a short window of initial response.[50,52] Thus, there is still an unmet need to deliver better alternative targeted therapies to NEPC patients.[51] Notably, Beltran et al.[38] proposed alisertib that targets aurora kinase A (AURKA) for overexpression of AURKA among NEPC tumors compared with both adenocarcinoma PC tumors and normal tissues. Though clinical evaluation of alisertib did not reach its primary end point, it suggested favorable results in biomarker-stratified subpopulations.[53] Corella et al.[15] adopted similar approaches to detect upregulated molecules in NEPC models and identified BCL-2 as an actionable target. Nonetheless, it still remains unclear whether differential expression analysis can pinpoint drug candidates with clinical impact. To date, there are few clinical trials that have tested interventions specifically in NEPC patients,[50] highlighting a need for new drugs to be evaluated in the clinic.

In recent years, computational methods have demonstrated utility in quickly repositioning existing drugs for specific indications in many cancers.[54,55] Computation-powered pipelines can offer drug candidates readily to be tested without the necessity of drug screening experiments. Here, we have designed a virtual drug screen pipeline to predict tumor response to various drugs in CRPC patient datasets. Inferred patient drug response was compared to NEPC molecular phenotype, through which drugs exhibiting strong predicted activity among tumors having high NEPC scores were further examined. Without the need to narrow down to specific targets through DE analysis, our pipelines discovered drug leads based on whole transcriptome profiles. Using this approach, we identified previously nominated drugs as well as novel drug classes. To evaluate proposed drugs and our computation rationale, we established an in vitro model system comprising an NEPC CCL (NCI-H660) and an Adeno-CRPC CCL (LNCaP). As a proof-of-concept, ABT-737, a BCL-2 inhibitor previously known to inhibit NEPC, induced growth inhibition at lower doses in NCI-H660 cells compared to LNCaP. In addition, an EZH2 inhibitor was also predicted to be efficacious against NEPC (Figure 2b). The epigenetic modifier EZH2 has been shown to facilitate NE differentiation under hormone therapy,[56,57] and suppressing this pathway has yielded inhibitory effects in NEPC tumors.[14,58] More importantly, we also observed a strong signal of NAMPT inhibitors that had not been investigated in NEPC before.

Abnormal expression of NAMPT has been shown in many cancers including pancreatic NE tumors,[40,42,43] and our in vitro experiments validated the preferential inhibitory effects of daporinad and GMX-1778 in NEPC cells compared with Adeno-CRPC cells. A previous phase I trial for daporinad established a tolerable dose of 0.126 mg/ $m^2$/h, corresponding to a mean plasma concentration of 5.51 (95% C.I.: 2.94–8.08) ng/mL.[59] Based on this, further calculation gives an estimated mean plasma concentration of 14.07 nM (95% C.I.: 7.51–20.64). In comparison, our measured daporinad $IC_{50}$ (approximately 0.54 nM) among NCI-H660 cells sits well below the recommended phase I tolerable dose, suggesting its ability to achieve strong anti-tumor effects with a low likelihood of inducing side effects. This further highlights the potential of daporinad to be evaluated in NEPC patients, either as a single-agent or in a drug combination to control aggressive heterogeneous CRPC tumors. In addition, recent independent work has reported favorable efficacy of NAMPT inhibitors in advanced prostate tumors and NE tumors including small cell lung cancer and NEPC,[60,61] expanding the potential therapeutic area of NAMPT inhibitors to other NE tumors. Taken together, the emerging evidence of promising effects of targeting NAMPT in NE tumors warrants clinical development of such a drug class. To this end, our identified sensitivity biomarkers can potentially expedite establishment of new treatments for NEPC.

Past therapeutic development experiences in NEPC have supported the importance of biomarkers to inform clinical decisions.[49] While attempts have been made to detect key molecules associated with NEPC,[62–64] most findings focused on delineation of NE differentiation from adenocarcinoma, instead of informing treatment outcomes among patients. To enable drug sensitivity biomarker discovery, we implemented causal inference approaches to detect direct causes and effects surrounding daporinad sensitivity. Compared with correlational models, causal feature selection algorithms test conditional independence and focus on variables with irremovable connections.[28,65] Our application of causal inference has elucidated a network containing key genes whose activities reflect daporinad sensitivity.

Based on the identified causal biomarkers, we constructed a signature set, namely DSS, to better inform vulnerabilities to NAMPT inhibitors. Validation by various means—including an independent CRPC patient biopsy scRNA-seq dataset—strongly supported our use of DSS to predict drug response. Moreover, genetic modifications of NAMPT and HNRNPA1 revised HCI-H660 sensitivity to daporinad in a manner that is consistent with expectations throughout. Thus far, our drug screens in gene manipulation conditions underscore adoptions of causal discovery methods for informing treatment response. We anticipate these identified biomarkers will assist in predicting patient subpopulations who will benefit from NAMPT inhibitors and facilitate effective clinical development of new treatment strategies. Nonetheless, the proposed biomarker discovery technique has limitations, largely related to properties of causal inference algorithms. First, it may require long computation times compared to traditional approaches, as many models are being fit to adjust for different variables. Also, the current settings require a pre-selection step for most differentially expressed genes, which may filter out some potential interesting markers. It may only pick out the ones that display consistent association with drug response, which could overlook genomic components belonging to the same biological pathway. However, paired with our experimental testing, we substantiated the use of such methods for detecting marker genes indicative of drug response.

In conclusion, we established a computational drug screen framework to quickly identify drug candidates with clinical impact for NEPC. Our approach quickly generated drugs that are ready to be tested without the need for large-scale experiments. Our drug candidates are supported by independent research. We incorporated causal inference methods to detect biomarkers indicative of drug response. Validation of the NAMPT sensitivity biomarkers substantiates our rationale and highlights their utility in clinical investigations of NAMPT inhibitors. We believe the proposed comprehensive workflow will amass interests from various fields to invigorate therapy development for different indications.

## AUTHOR CONTRIBUTIONS

W.Z., A.L., S.M.D., and R.S.H. wrote the manuscript; W.Z., A.L., and R.S.H. designed the research; W.Z., A.L., and L.L. performed the research; W.Z. and A.L. analyzed the data; W.Z. and A.L. contributed new reagents/analytical tools.

## FUNDING INFORMATION

This study was supported by NIH/NCI Grants R01CA204856 (R.S.H) and the University of Minnesota (UMN) Office of Academic Clinical Affairs (OACA) Faculty Research Development grant (R.S.H and S.D.) and UMN College of Pharmacy SURRGE award (R.S.H). R.S.H. also received support from NIH/NCI R01CA229618, a UMN OACA Grant-in-Aid Program (GIA) award. W.Z. received the UMN Bioinformatics and Computational Biology first-year Fellowship, the UMN IDF Fellowship, the UMN DDF Fellowship, and the UMN Clinical & Translational Science Institute A-PReP scholarship.

## CONFLICT OF INTEREST STATEMENT

The authors declare no competing interests for this work.

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

**How to cite this article:** Zhang W, Lee A, Lee L, Dehm SM, Huang RS. Computational drug discovery pipelines identify NAMPT as a therapeutic target in neuroendocrine prostate cancer. *Clin Transl Sci*. 2024;17:e70030. doi:10.1111/cts.70030